

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{int} \times \text{int} \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{int} \Rightarrow \text{Process} \\ \text{sem} : \text{int} \times \text{int} \times \text{int} \Rightarrow \text{Semaphore} \\ \text{rdr7} : \text{Fcall} \\ \text{rdr9} : \text{int} \Rightarrow \text{Fcall} \\ \text{rdr10} : \text{int} \Rightarrow \text{Fcall} \\ \text{rdr11} : \text{int} \Rightarrow \text{Fcall} \\ \text{wrtr18} : \text{Fcall} \\ \text{wrtr19} : \text{Fcall} \\ \text{wrtr20} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1:	$\text{cnfg}(\text{p}(\text{rdr7}, 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr9}(0), 0), p2, p3, \text{sem}(s, d, t), x, 0)$	
2:	$\text{cnfg}(\text{p}(\text{rdr9}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr10}(y), 0), p2, p3, \text{sem}(s1, d, t), x, 0)$	$[\neg s = 0 \wedge s1 = s - 1]$
3:	$\text{cnfg}(\text{p}(\text{rdr9}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr9}(y), t), p2, p3, \text{sem}(s, d, t2), x, 0)$	$[s = 0 \wedge t2 = t + 2]$
4:	$\text{cnfg}(\text{p}(\text{rdr9}(y), n), p2, p3, \text{sem}(s, d, t), x, 1) \rightarrow \text{cnfg}(\text{p}(\text{rdr10}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0)$	$[n = d \wedge \neg n = 0]$
5:	$\text{cnfg}(\text{p}(\text{rdr10}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr11}(x), 0), p2, p3, \text{sem}(s, d, t), x, 0)$	
6:	$\text{cnfg}(\text{p}(\text{rdr11}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr7}, 0), p2, p3, \text{sem}(s, d2, t), x, 1)$	$[\neg t = d + 2 \wedge d2 = d + 2]$
7:	$\text{cnfg}(\text{p}(\text{rdr11}(y), 0), p2, p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(\text{p}(\text{rdr7}, 0), p2, p3, \text{sem}(s1, d2, t), x, 0)$	$[t = d + 2 \wedge s1 = s + 1]$
8:	$\text{cnfg}(p1, \text{p}(\text{rdr7}, 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr9}(0), 0), p3, \text{sem}(s, d, t), x, 0)$	
9:	$\text{cnfg}(p1, \text{p}(\text{rdr9}(y), 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr10}(y), 0), p3, \text{sem}(s1, d, t), x, 0)$	$[\neg s = 0 \wedge s1 = s - 1]$
10:	$\text{cnfg}(p1, \text{p}(\text{rdr9}(y), 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr9}(y), t), p3, \text{sem}(s, d, t2), x, 0)$	$[s = 0 \wedge t2 = t + 2]$
11:	$\text{cnfg}(p1, \text{p}(\text{rdr9}(y), n), p3, \text{sem}(s, d, t), x, 1) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr10}(y), 0), p3, \text{sem}(s, d, t), x, 0)$	$[n = d \wedge \neg n = 0]$
12:	$\text{cnfg}(p1, \text{p}(\text{rdr10}(y), 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr11}(x), 0), p3, \text{sem}(s, d, t), x, 0)$	
13:	$\text{cnfg}(p1, \text{p}(\text{rdr11}(y), 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr7}, 0), p3, \text{sem}(s, d2, t), x, 1)$	$[\neg t = d + 2 \wedge d2 = d + 2]$
14:	$\text{cnfg}(p1, \text{p}(\text{rdr11}(y), 0), p3, \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, \text{p}(\text{rdr7}, 0), p3, \text{sem}(s1, d2, t), x, 0)$	$[t = d + 2 \wedge s1 = s + 1]$
15:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, 0), \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr19}, 0), \text{sem}(s1, d, t), x, 0)$	$[\neg s = 0 \wedge s1 = s - 1]$
16:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, 0), \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, t), \text{sem}(s, d, t2), x, 0)$	$[s = 0 \wedge t2 = t + 2]$
17:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, n), \text{sem}(s, d, t), x, 1) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr19}, 0), \text{sem}(s, d, t), x, 0)$	$[n = d \wedge \neg n = 0]$
18:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr19}, 0), \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr20}, 0), \text{sem}(s, d, t), 1, 0)$	
19:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr20}, 0), \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, 0), \text{sem}(s, d2, t), x, 1)$	$[\neg t = d + 2 \wedge d2 = d + 2]$
20:	$\text{cnfg}(p1, p2, \text{p}(\text{wrtr20}, 0), \text{sem}(s, d, t), x, 0) \rightarrow \text{cnfg}(p1, p2, \text{p}(\text{wrtr18}, 0), \text{sem}(s1, d, t), x, 0)$	$[t = d + 2 \wedge s1 = s + 1]$
21:	$\text{cnfg}(p1, p2, p3, \text{sem}(s, d, t), x, a) \rightarrow \text{success}$	
22:	$\text{cnfg}(\text{p}(\text{rdr10}(y), n1), \text{p}(\text{rdr10}(y), n2), \text{p}(\text{wrtr19}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
23:	$\text{cnfg}(\text{p}(\text{rdr11}(y), n1), \text{p}(\text{rdr10}(y), n2), \text{p}(\text{wrtr19}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
24:	$\text{cnfg}(\text{p}(\text{rdr10}(y), n1), \text{p}(\text{rdr11}(y), n2), \text{p}(\text{wrtr19}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
25:	$\text{cnfg}(\text{p}(\text{rdr11}(y), n1), \text{p}(\text{rdr11}(y), n2), \text{p}(\text{wrtr19}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
26:	$\text{cnfg}(\text{p}(\text{rdr10}(y), n1), \text{p}(\text{rdr10}(y), n2), \text{p}(\text{wrtr20}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
27:	$\text{cnfg}(\text{p}(\text{rdr11}(y), n1), \text{p}(\text{rdr10}(y), n2), \text{p}(\text{wrtr20}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
28:	$\text{cnfg}(\text{p}(\text{rdr10}(y), n1), \text{p}(\text{rdr11}(y), n2), \text{p}(\text{wrtr20}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	
29:	$\text{cnfg}(\text{p}(\text{rdr11}(y), n1), \text{p}(\text{rdr11}(y), n2), \text{p}(\text{wrtr20}, n3), \text{sem}(s, d, t), x, a) \rightarrow \text{error}$	

$$\{30 : \text{cnfg}(p(\text{rd}r7, 0), p(\text{rd}r7, 0), p(\text{vrt}r18, 0), \text{sen}(2, d, t), 0, 0) \Rightarrow \text{succss}[(d > 0) \wedge t == d + 2]\}$$

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{bitvec}_{32} \times \text{bitvec}_1 \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{bitvec}_3 \Rightarrow \text{Process} \\ \text{sem} : \text{bitvec}_2 \times \text{bitvec}_3 \times \text{bitvec}_3 \Rightarrow \text{Semaphore} \\ \text{rdr7} : \text{Fcall} \\ \text{rdr9} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{rdr10} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{rdr11} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{wrtr18} : \text{Fcall} \\ \text{wrtr19} : \text{Fcall} \\ \text{wrtr20} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1:	cnfg(p(rdr7, #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr9(#x00000000), #b000), p2, p3, sem(s, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
2:	cnfg(p(rdr9(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr10(y), #b000), p2, p3, sem(s1, d, t), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
3:	cnfg(p(rdr9(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr9(y), t), p2, p3, sem(s, d, t2), x, #b0)	[n = d ∧ ¬n = #b000]
4:	cnfg(p(rdr9(y), n), p2, p3, sem(s, d, t), x, #b1) → cnfg(p(rdr10(y), #b000), p2, p3, sem(s, d, t), x, #b0)	
5:	cnfg(p(rdr10(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr11(x), #b000), p2, p3, sem(s, d, t), x, #b0)	
6:	cnfg(p(rdr11(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr7, #b000), p2, p3, sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
7:	cnfg(p(rdr11(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr7, #b000), p2, p3, sem(s1, d2, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
8:	cnfg(p1, p(rdr7, #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr9(#x00000000), #b000), p3, sem(s, d, t), x, #b0)	
9:	cnfg(p1, p(rdr9(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr10(y), #b000), p3, sem(s1, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
10:	cnfg(p1, p(rdr9(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr9(y), t), p3, sem(s, d, t2), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
11:	cnfg(p1, p(rdr9(y), n), p3, sem(s, d, t), x, #b1) → cnfg(p1, p(rdr10(y), #b000), p3, sem(s, d, t), x, #b0)	[n = d ∧ ¬n = #b000]
12:	cnfg(p1, p(rdr10(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr11(x), #b000), p3, sem(s, d, t), x, #b0)	
13:	cnfg(p1, p(rdr11(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr7, #b000), p3, sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
14:	cnfg(p1, p(rdr11(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr7, #b000), p3, sem(s1, d2, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
15:	cnfg(p1, p2, p(wrtr18, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr19, #b000), sem(s1, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
16:	cnfg(p1, p2, p(wrtr18, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, t), sem(s, d, t2), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
17:	cnfg(p1, p2, p(wrtr18, n), sem(s, d, t), x, #b1) → cnfg(p1, p2, p(wrtr19, #b000), sem(s, d, t), x, #b0)	[n = d ∧ ¬n = #b000]
18:	cnfg(p1, p2, p(wrtr19, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), #x00000001, #b0)	
19:	cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, #b000), sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
20:	cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, #b000), sem(s1, d, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
21:	cnfg(p1, p2, p3, sem(s, d, t), x, a) → success	
22:	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
23:	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
24:	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
25:	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
26:	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
27:	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
28:	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
29:	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	

$$\{ \exists d \text{ cnfg}(p(\text{rd}r7, \#b000), p(\text{rd}r7, \#b000), p(\text{wrt}r18, \#b000), \text{sen}(\#b10, d, t), \#x00000000, \#b0) \Rightarrow \text{succes}[(d \vee \text{ugt} \#b000) \wedge t = d \vee \text{add} \#b010] \}$$